

Technical Requirements Document (TRD) — FindLostPuppy

| Document Metadata | Value | | :--- | | :--- | | **System** | FindLostPuppy Multi-Platform Client & Cloud Architecture | | **Document Version** | 1.0.0 | | **Frontend Framework** | React 19, TypeScript 6.0, Vite 8.2 | | **Mobile Bridge** | Capacitor 8.5 (Android AGP 8.3+, Gradle 8.7+, Java 17) | | **Cloud Services** | Google Cloud Firestore, Firebase Auth, Firebase Storage, Cloudinary REST API | | **Security Standard** | Deny-by-Default, Zero-PII-Leak, Client-Side Image Compression |

1. System Architecture Overview

CLIENT APPLICATION | | React 19 + TypeScript +

Vite Bundler | | Capacitor Android Native Container

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Firebase Auth | | Cloud Firestore | | Cloudinary

Media | | Google OAuth / | | Real-time DB & | | Unsigned Public | | Session Bridge | | Security Rules | | Alert Uploads

Firebase Storage | | Private Owner / | | Pet Media Storage

2. Frontend Architecture & State Management

2.1 Technology Stack Matrix

- **Runtime & UI:** React 19 (`react` , `react-dom`) with TypeScript (`~6.0.2`).
- **Routing:** React Router v7 (`react-router-dom`) with SPA deep link fallback handling (`public/404.html` `$\rightarrow` `index.html` `window.history.replaceState`).
- **Styling:** Custom Obsidian Titanium design system (`src/index.css` , `src/App.css`) with Tailwind utility integration in `src/components/ui/` (`tailwind-merge` , `clsx` , `class-variance-authority`).
- **Build System:** Vite 8.2 with Rolldown-ready code splitting and chunk size optimization.

- **Native Container:** @capacitor/core , @capacitor/android , @capacitor/geolocation , @capacitor-firebase/authentication .

2.2 Local Storage Schema & Reactive Events

The application employs an offline-first storage repository pattern

(src/services/storageService.ts):

LocalStorage Key	Schema Type	Description
findlostpuppy_active_user	User object	Current authenticated session (ID, name, email, phone).
findlostpuppy_profiles_v1	OwnerProfile[]	Local array of saved owner profiles.
findlostpuppy_pets_v1	DogProfile[]	Local array of registered pet profiles.
findlostpuppy_reports_v1	LostReport[]	Local cache of missing and safe pet reports.
findlostpuppy_sightings_v1	Sighting[]	Local cache of community sightings.
findlostpuppy_consent_v1	ConsentRecord	Affirmative terms & privacy acceptance version stamp.

Cross-Component Reactivity: Components synchronize state via browser custom events:

- findlostpuppy_reports_updated : Dispatched whenever a report or sighting is added/updated.
- findlostpuppy_data_synced : Dispatched on successful Firestore cloud sync.
- storage : Dispatched on direct localStorage writes to keep active tabs in sync.

3. Data Models & TypeScript Interfaces

3.1 Core Data Interfaces (src/types/index.ts)

```
typescript export interface DogProfile { id: string; ownerId: string; name: string; breed: string; color: string; size: 'Small' | 'Medium' | 'Large' | 'Extra Large'; gender: 'Male' | 'Female'; age?: string; distinguishingMarks?: string; hasCollarOrChip?: boolean; collarDetails?: string; primaryPhoto: string; photos: string[]; createdAt?: string; updatedAt?: string; }
```

```
export interface OwnerProfile { id: string; userId: string; fullName: string; phone: string; email: string; preferredContact: 'phone' | 'email' | 'whatsapp'; state: string; district: string; mandalOrMunicipality: string; city: string; // Village / Locality / Home Base streetOrLocality?: string; latitude?: number; longitude?: number; hasLocationConsent: boolean; photoUrl?: string; updatedAt: string; }
```

```
export interface LostReport { id: string; dogId: string; ownerId: string; status: 'LOST' | 'SAFE'; dateLost: string; timeLost: string; lastKnownLocation: string; ownerApproximateLocation: string; // Masked locality (State, District, Mandal, Village) dog: DogProfile; sightingCount: number; contactMechanism: { showPhone: boolean; showEmail: boolean; safeContactPhone?: string; // MASKED (e.g. +91 86.....48) safeContactEmail?: string; // MASKED (e.g. j...5@gmail.com) }; createdAt: string; updatedAt: string; }
```

```
export interface Sighting { id: string; reportId: string; dogName: string; date: string; time: string; location: string; state: string; district: string; mandal: string; village:
```

```
string; photo: string; description: string; latitude?: number; longitude?: number; isCurrent:
boolean; createdAt: string; }
```

4. Cloud & Backend Specifications

4.1 Cloud Firestore Collections & Security Rules (`firebase.firestore.rules`)

All Firestore access is `deny-by-default`.

Collection	Read Permission	Write / Update Permission	Privacy Invariant
<code>owner_profiles</code>	Authenticated Owner / Admin	Authenticated Owner / Admin	Exact coordinates and unmasked phone never exposed publicly.
<code>pets</code>	Authenticated Owner / Admin	Authenticated Owner / Admin	Private pet profile data.
<code>missing_reports</code>	Public Authenticated Read (Sanitized)	Authenticated Owner / Admin	Sanitized payload: no exact coordinates, masked contacts only.
<code>sightings</code>	Public Authenticated Read (Sanitized)	Public Authenticated Write	Image uploads sanitized; no PII embedded.
<code>safe_reports</code>	Authenticated Owner / Admin	Authenticated Owner / Admin	Private safe confirmation.
<code>user_feedback</code>	Authenticated User	Authenticated User	Bounded text feedback for community rescue enhancements.

4.2 Cloudinary Media Integration

- **Unsigned Upload Preset:** Restricted solely to public missing pet photos and sighting reports under the directory `findlostpuppy/public-alerts/`.
- **Private Profiles:** Owner personal photos and safe pet photos are stored in Firebase Storage with access rules locked to the owning user UID.
- **Client-Side Compression:** Images are compressed via `compressImage` (HTML5 Canvas rendering, max width/height `1280px` , JPEG quality `0.75`) prior to transmission.

5. Security, Privacy & Integrity Specifications

5.1 Indian Phone Validation

```
Phone fields must strictly satisfy validateIndianPhoneNumber : `typescript export function
validateIndianPhoneNumber(phone: string): boolean { const digits = phone.replace(/\D/g, '');
return digits.length === 10 && /^[6-9]/.test(digits); } `
```

5.2 Zero-PII Leak Masking Algorithm

```
`typescript export function maskPhoneNumber(phone: string): string { const digits =
phone.replace(/\D/g, ''); if (digits.length !== 10) return '+91 .....'; return '+91
${digits.slice(0, 2)}.....${digits.slice(-2)} ; }
```

```
export function maskEmailAddress(email: string): string { const [user, domain] =
email.split('@'); if (!user || !domain) return '...@....com'; const maskedUser = user.length
```

```
≤ 2 ? ${user[0]}• : ${user[0]}•••${user.slice(-1)} ; return ${maskedUser}@${domain} ; }
```

5.3 Android App Backup Invariant

In `android/app/src/main/AndroidManifest.xml` : `xmlns:android="http://schemas.android.com/apk/res/android"` Prevents unauthorized local storage extraction of credentials or private pet documents via Android Debug Bridge (`adb backup`).

6. Build, Android Packaging & Release Automation

6.1 Release Validation Command Chain

Before any code deployment or Android build generation, the following validation chain must execute with 0 errors:

```
bash npm run lint # OXlint validation across codebase npm run build # TypeScript typecheck (tsc -b) & Vite compilation npm run build:android # Validates env and builds with CAPACITOR_BUILD=true npm run sync:android # Capacitor assets copy & Android verify
```

6.2 Android AAB Release Deliverables

When compiling a production Android App Bundle (AAB):

- Outputs must be generated via Gradle: `./gradlew bundleRelease`
- Copies must be deposited on the Desktop under both naming standards:

1. `findlostpuppy-release.aab` 2. `findlostpuppy-v{versionName}-code{versionCode}-release.aab`